

Lesson 18 Equivalence Relations

1. For each of the following, determine if it is an equivalence relation. If it is not, explain why not. If it is, give its equivalence classes.
 - a) $R = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} : y = 2x\}$
 - b) $R = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} : |x - y| \leq 2\}$
 - c) R is a relation on $A = \{x \in \mathbb{N} : x < 100\}$ and xRy if x and y have the same “ones” digit.
 - d) R is a relation on $\{a, b, c\}$,
 $R = \{(a, a), (b, b), (c, c), (a, c), (c, a), (b, c), (c, b)\}$
 - e) R is a relation on $\mathbb{Z}$, such that xRy if and only if $xy \geq 0$.
2. Let R be the congruence modulo 2 relation on $\mathbb{Z}$. Show that $[a] = [-a]$ for every $a \in \mathbb{Z}$.
3. Given that R is an equivalence relation on A and the equivalence classes of R are $[1] = \{1, 5, 6\}$, $[2] = \{2, 4\}$, and $[3] = \{3\}$, write out the sets A and R .
4. Let $A = \{0, 1, 2, \dots, 10\}$ and $R = \{(a, b) \in A \times A : 2|(a + b)\}$.
 - a) Show that R is an equivalence relation.
 - b) Compute the distinct equivalence classes of R .
5. Let $R = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} : x^2 = y^2\}$. Show that R is an equivalence relation and compute the distinct equivalence classes of R .
6. Let R be a relation on $A = \mathbb{Z} - \{0\}$, such that xRy if and only if $xy > 0$. Show that R is an equivalence relation and compute the distinct equivalence classes of R .
7. Let R be a relation that defines equivalent fractions on $\mathbb{Q} = \{m/n : m, n \in \mathbb{Z}, n \neq 0\}$,

$$\left(\frac{a}{b}, \frac{c}{d}\right) \in R \quad \text{if } ad = bc$$

Show that R is an equivalence relation.

8. Consider an equivalence relation on set A . Show that $a \in [b]$ if and only if $b \in [a]$ for all $a, b \in A$.